

Car Seat Sales Forecasting Using Random Forest

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Executive Summary

This analysis applies a **Random Forest regression model** to forecast car seat sales across retail stores using demographic, pricing, and merchandising attributes.

Random Forest extends the baseline CART model by leveraging an **ensemble of decision trees**, resulting in improved predictive accuracy and robustness to noise and outliers. The model is trained using **5-fold cross-validation** and evaluated using consistent performance metrics to ensure fair comparison across modeling approaches.

The trained model is applied to **new, unseen store data** to generate forward-looking sales forecasts that can support:

- Inventory and demand planning
- Store-level pricing strategy
- Advertising budget optimization
- Risk mitigation for underperforming store configurations

This Random Forest model serves as a **high-performing benchmark**, balancing predictive accuracy and scalability, while complementing the interpretability of CART and the flexibility of Neural Networks.

1. Project Overview

Accurate sales forecasting enables retailers to optimize inventory levels, pricing strategies, and marketing investments.

This notebook implements a **Random Forest regression model** to forecast car seat sales using store-level demographic, pricing, and merchandising attributes. The analysis follows the same data pipeline, validation strategy, and evaluation framework as other models in this project to ensure fair and consistent comparison.

2. Business Objective

The primary objective of this analysis is to:

- Forecast car seat sales across retail stores
 - Improve prediction accuracy beyond a single decision tree
 - Capture non-linear patterns and feature interactions
 - Generate reliable sales estimates for new, unseen stores
-

3. Dataset Overview

The dataset contains store-level attributes related to competition, pricing, demographics, and merchandising.

Target Variable

- **Sales** – Number of car seats sold (measured in hundreds)

Predictor Variables

- **CompPrice** – Competitor pricing
- **Income** – Local average income (in \$000s)
- **Advertising** – Advertising spend (in \$000s)
- **Population** – Local population size (in \$000s)
- **Price** – Store price
- **ShelfLoc** – Shelf location quality
- **Age** – Average customer age
- **Education** – Average education level
- **Urban** – Urban location indicator
- **US** – U.S. store indicator

The variable **Store** is excluded from modeling as it is a unique identifier.

4. Environment Setup

```
library(caret)
library(randomForest)
library(dplyr)
```

5. Data Ingestion

The dataset is loaded from the local project directory. This dataset contains historical sales and store-level attributes used for model training.

```
car_seat_sales <- read.csv("data/car_seat_sales.csv")
```

6. Data Preparation

The `Store` variable is removed from the modeling dataset, as it serves only as a unique identifier and does not contribute to predictive performance.

```
library(dplyr)

model_data <- car_seat_sales %>%
  dplyr::select(-Store)
```

7. Modeling Strategy

Why Random Forest?

Random Forest is an ensemble learning method that aggregates predictions from multiple decision trees to improve accuracy and generalization.

Key advantages include:

- Reduced overfitting compared to single trees
- Improved predictive performance
- Robust handling of non-linear relationships
- Implicit modeling of feature interactions

Validation Approach

A **5-fold cross-validation** strategy is used to ensure model stability and enable fair comparison with other models in this project.

8. Hyperparameter Tuning Strategy

The primary hyperparameter in Random Forest regression is `mtry`, which controls the number of candidate features considered at each split.

```
set.seed(1975)

library(caret)

cv_method <- trainControl(
  method = "cv",
  number = 5
)
```

```
mtry_grid <- expand.grid(
  .mtry = 1:10
)
```

Tuning the *mtry* parameter allows the model to balance randomness and predictive strength by controlling how many features compete at each split.

9. Model Training – Random Forest

The Random Forest model is trained using the defined cross-validation strategy and hyperparameter grid.

```
rf_model <- train(
  Sales ~ .,
  data = model_data,
  method = "rf",
  trControl = cv_method,
  tuneGrid = mtry_grid
)
```

```
rf_model
```

```
## Random Forest
##
## 400 samples
## 10 predictor
##
## No pre-processing
## Resampling: Cross-Validated (5 fold)
## Summary of sample sizes: 321, 319, 321, 320, 319
## Resampling results across tuning parameters:
##
##   mtry  RMSE      Rsquared    MAE
##   1     5.134912  0.007074219  4.422608
##   2     5.213840  0.004745531  4.473506
##   3     5.253911  0.003705958  4.503423
##   4     5.303549  0.009648952  4.544980
##   5     5.299385  0.009462189  4.529780
##   6     5.326357  0.009716073  4.551656
##   7     5.326233  0.007453452  4.555771
##   8     5.336907  0.007433454  4.559910
##   9     5.324815  0.007868152  4.554085
##  10     5.351192  0.012187310  4.578438
##
## RMSE was used to select the optimal model using the smallest value.
## The final value used for the model was mtry = 1.
```

10. Model Evaluation

The trained model object contains cross-validated performance metrics and the optimal hyperparameter configuration.

```
rf_model$bestTune
```

```
##      mtry  
## 1      1
```

11. Generating Predictions on New Store Data

In a production setting, the trained model is applied to new store configurations where historical sales data is unavailable.

Load New Data

```
new_car_seat_sales <- read.csv("data/new_car_seat_sales_data.csv")  
  
new_model_data <- new_car_seat_sales %>%  
  dplyr::select(-Store)
```

Generate Predictions

```
rf_predictions <- predict(rf_model, new_model_data)  
  
prediction_results <- new_car_seat_sales %>%  
  mutate(Predicted_Sales = rf_predictions)  
  
head(prediction_results)
```

```
##      Store CompPrice Income Advertising Population Price ShelfLoc Age Education  
## 1      216      116      83          15         170   144      Bad   71         11  
## 2        8      136      81          15         425   120      Good   67         10  
## 3      273      113      33           0          14    63      Good   38         12  
##      Urban  US Predicted_Sales  
## 1    Yes Yes          9.744427  
## 2    Yes Yes          8.107728  
## 3    Yes No          6.303432
```

prediction_results represents the expected number of car seats sold (measured in hundreds of units) for each new store.

12. Business Insights

The Random Forest model provides improved forecasting accuracy by aggregating multiple decision trees.

This approach delivers:

- More stable predictions
 - Reduced sensitivity to individual data points
 - Strong performance in complex, non-linear environments
-

13. Limitations and Next Steps

Final model selection will be guided by cross-validated performance metrics and business requirements such as interpretability and deployment constraints.